



SDAIA

الهيئة السعودية للبيانات
والذكاء الاصطناعي
Saudi Data & AI Authority

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Artificial Intelligence

Artificial Intelligence

Artificial Intelligence (AI) is one of the cutting-edge technologies that significantly contributes to rapid technological development and increases opportunities for innovation and growth in various fields. AI plays a vital role in enhancing quality, increasing capabilities, business efficiency, and improving productivity. Despite the widespread use of AI technologies and the much-talked-about AI capabilities, there is some uncertainty about these technologies, which could lead to unrealistic expectations. This makes it difficult for many decision-makers or executives in the public and private sectors to comprehend AI, AI technologies, and the reality of AI capabilities.



What is AI?

AI is systems that use technologies capable of making predictions, generating content, providing recommendations, or making decisions with different levels of autonomy.

History

The first proposals for an artificial neuron model appeared at the beginning of the 1940s, which is when artificial intelligence first emerged. When British scientist Alan Turing questioned whether a machine could think, the idea of artificial intelligence started taking off in the early 1950s. Since then, AI has experienced waves of growth and decline, known as "AI Winter," before it gained the widespread adoption that we are seeing today in many industries. The most important advancements in the development of AI capabilities are depicted in Figure 1 as a timeline.

AI Objectives



Enhancing business efficiency, quality, and productivity.



Making smart decisions and providing accurate recommendations.



Automating tasks to save time and costs.



Supporting humans in decision-making and data analysis.



Creating new content such as texts, images, and videos.

National Artificial Intelligence Index (NAII)

The National Artificial Intelligence Index (NAII) aims to assess the readiness of government entities to adopt artificial intelligence (AI) technologies, monitor their progress, and provide tailored recommendations that enhance their ability to develop and implement effective and sustainable AI solutions. It serves as a strategic tool to support the goals of Saudi Vision 2030 by accelerating the adoption of AI across sectors.

For more, see the National Artificial Intelligence Index: [National Artificial Intelligence Index \(NAII\)](#) ↗

AI Adoption Framework

The AI adoption framework is a guiding reference that provides a comprehensive

The AI adoption framework is a guiding reference that provides a comprehensive approach for AI adoption across all sectors. It aims to establish directions, guidelines, and identify key steps in according with best practices. The AI adoption framework helps ensure achievement of desired goals through the optimal and responsible use of AI.

For more, see the AI Adoption Framework: [AI Adoption Framework](#) 

Principles and Controls of AI Ethics

Given the rapid growth of practices and technologies in the field of AI and the diversity seen in the use of various applications in many fields, which has an effective role in accelerating the pace of decision-making and its optimal use, the principles and controls of AI ethics contribute to the application of ethics during the stages of the development lifecycle of AI systems. These principles also help support initiatives of research, development and innovation in the Kingdom, which is reflected in the quality of services provided by the Kingdom to individuals to ensure the responsible use of AI technologies.

For more, see the draft of Principles of AI Ethics: [Principles of AI Ethics](#) 

AI Principles in Media

Guiding principles governing the use of artificial intelligence across the media content lifecycle, ensuring the responsible and safe use of AI technologies in media.

For more, see the draft of AI Principles in Media : [AI Principles in Media](#) 

National Occupational Standard Framework for Data & Artificial Intelligence

This document highlights the role of the Saudi Data and AI Authority (SDAIA) as the national reference body for regulating the data and AI sector, through the issuance of a national framework to unify practices and enhance human capabilities.

For more, see the National Occupational Standard Framework for Data & Artificial Intelligence: [National Occupational Standard Framework for Data & Artificial Intelligence](#) 

Saudi Academic Framework for AI Qualifications (Education Intelligence)

This document presents the Saudi Academic Framework for Artificial Intelligence by SDAIA. It serves as the official guide for the development and accreditation of higher education programs and is a trusted and recognized reference in Saudi Arabia.

For more, see the Saudi Academic Framework for AI Qualifications (Education Intelligence): [Saudi Academic Framework for AI Qualifications \(Education Intelligence\)](#) 

Global Figures and Statistics



\$19.9 Trillion

of AI's contribution to the global economy by 2030.



\$4.8 Trillion

is the market value of AI in 2033.



170 Million jobs

will be created because of AI by 2030.

Generative AI (GenAI)

In 2022, the world witnessed a new technological revolution, resulting in the emergence of Generative AI technologies that create creative content, such as texts, videos, and audio. This comes as a result of the growing global interest over

the past decade in the rapid development of modern technologies, especially in AI. These technologies have spread widely due to their capabilities and broad horizons for utilization and innovation in various fields, unlike previous models that performed only one task. Therefore, these technologies have contributed to improving services, increasing productivity, and boosting revenues.

What is Generative AI?

The **Generative AI** is a branch of artificial intelligence that relies on machine learning techniques and deep neural networks to simulate the human ability to produce original data and content.

Generative AI Guideline

There are two versions of this guideline. The first version directed toward government employees. The second version is for the public. Both versions provide guidelines regarding adoption and use of generative artificial intelligence systems. It also include examples based on common scenarios that entities may address. It also highlights the challenges and considerations associated with the use of generative artificial intelligence, proposes principles for responsible use, and presents recommended practices.

For more, see the Generative AI Guidelines:

- ✓ **Generative AI Guideline for Government** [↗](#)
- ✓ **Generative AI Guideline for Public** [↗](#)

Deepfakes mitigating risks while fostering innovation

This document outlines the challenges associated with deepfake technologies in artificial intelligence and the resulting risks related to misinformation and privacy violations, particularly amid the rapid expansion of AI applications and their diverse uses across various sectors. It also presents key response tools and recommendations to mitigate these risks, emphasizing the importance of raising awareness, building trust, and ensuring the responsible use of such technologies in a manner that supports innovation and contributes to a safe environment for these

Global Figures and Statistics



\$2.6 - 4.4 Trillion

is the annual contribution of GenAI to the global economy.



\$1.3 Trillion

is the value of the GenAI market by 2032.



25%

Million jobs worldwide will be affected by GenAI.

Agentic AI

AI era is witnessing a fundamental shift from generative models concerned with content production to **Agentic AI** technologies capable of planning and self-executing complex tasks without human intervention or with limited human interventions. These new systems represent a qualitative leap from the concept of smart assistants to autonomous AI agents, as the role of AI is no longer limited to responding to commands, but has expanded to include executing actions, making decisions, and interacting with changing environments. Therefore, AI's new role contributed to opening broad horizons in automating processes, improving efficiency, and driving institutional and national innovation toward new horizons.

What is Agentic AI?

Agentic AI is a software system based on AI algorithms, characterized by key features such as perceiving the surrounding environment, interpreting information, setting goals, and making decisions autonomously without constant human supervision.

Global Figures and Statistics



\$450 Billion

of economic value is predicted to come from AI agents by 2028.



\$50 Billion

is the global Agentic AI market size by 2030.



40%

of knowledge-based tasks will be affected through Agentic AI by 2027.

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